DE:CalVal

An Integrated Approach from Local Observations to Forest Carbon Modelling

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Technische Universität Dresden
Wilderness International
GFZ Helmholtz Centre for Geosciences
UFZ HELMHOLTZ Centre for Environmental Research
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Team



Max Planck Institute for Biogeochemistry



University Maryland



Technical University Dresden



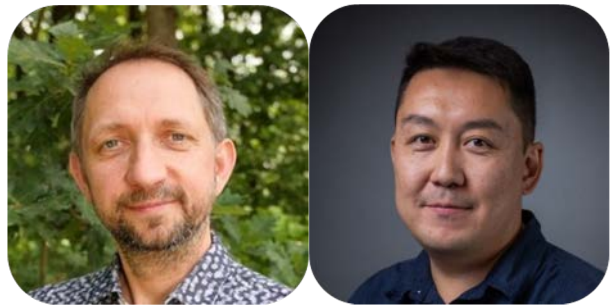
DLR



UFZ



Wilderness International

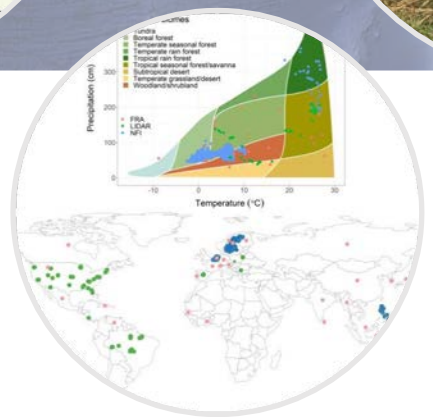


GFZ Potsdam


Cuba
Puerto Rico
Caribbean Sea
Caracas
Venezuela
Guyana
Suriname
French Guiana
Manaus
AMAZONAS
Brazil
Peru
Lima
Bolivia
La Paz
Curitiba
Mauritania
Mali
Niger
Senegal
Guinea-Bissau
Guinea
Burkina Faso
Sierra Leone
Liberia
Suden
GEO-TREES concept
• ALS sampled area 1,000 ha
• Forest inventory 10 ha
• TLS 3 ha
AIRBORNE LASER SCANNING (ALS)
FOREST PLOT INVENTORY


GEDI
ECOSYSTEM LIDAR


esa
biomass

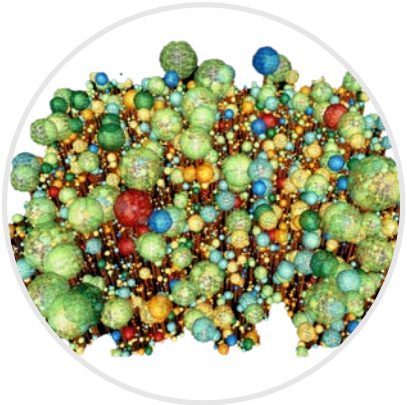

Precipitation (cm)
Temperature (°C)

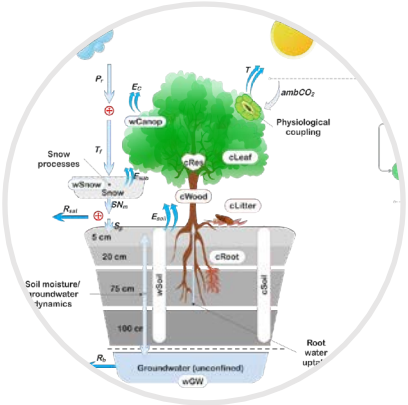

TanDEM


FLUXNET
A Global Network


SAPFLUXNET


GEO TREES




Physiological coupling
Snow processes
Snow
Soil moisture/ groundwater dynamics
Root water uptake
Groundwater (unconfined)

Focus #1 Campaigns

Focus #2 Reference data

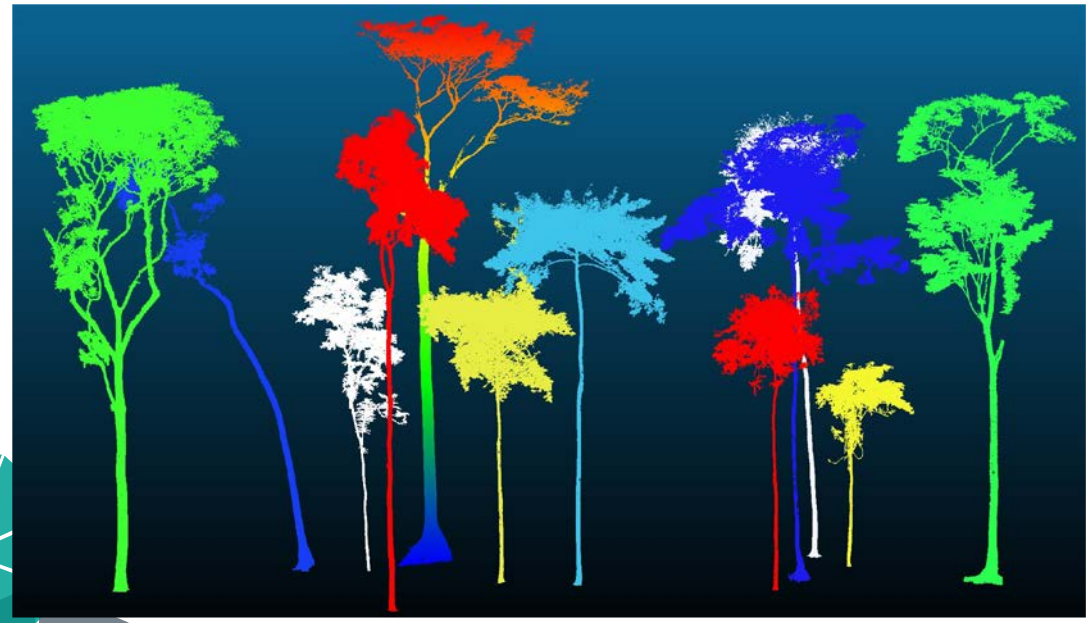
Focus #3 Carbon cycle

Focus #1: In situ campaigns

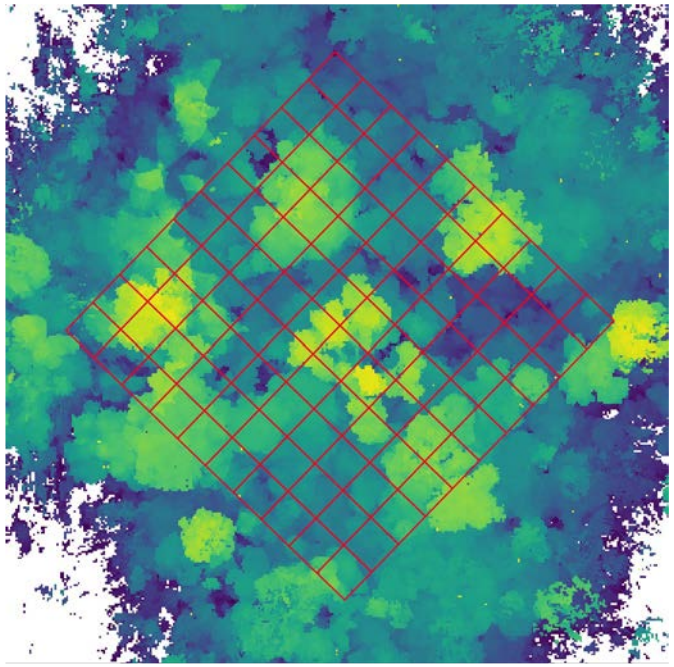
TLS campaigns to support BIOMASS cal/val activities

- ***completed***: Ghana (January 2026) and Gabon (September 2026)
- **planned**: French Guiana (Sep-Oct 2027)

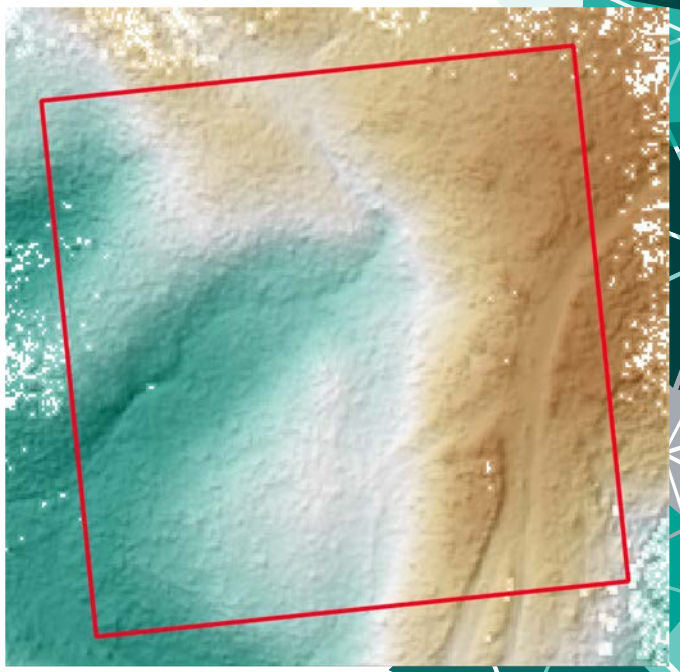
Individual tree segmentation in Ankasa, Ghana



CHM in Ipassa, Gabon

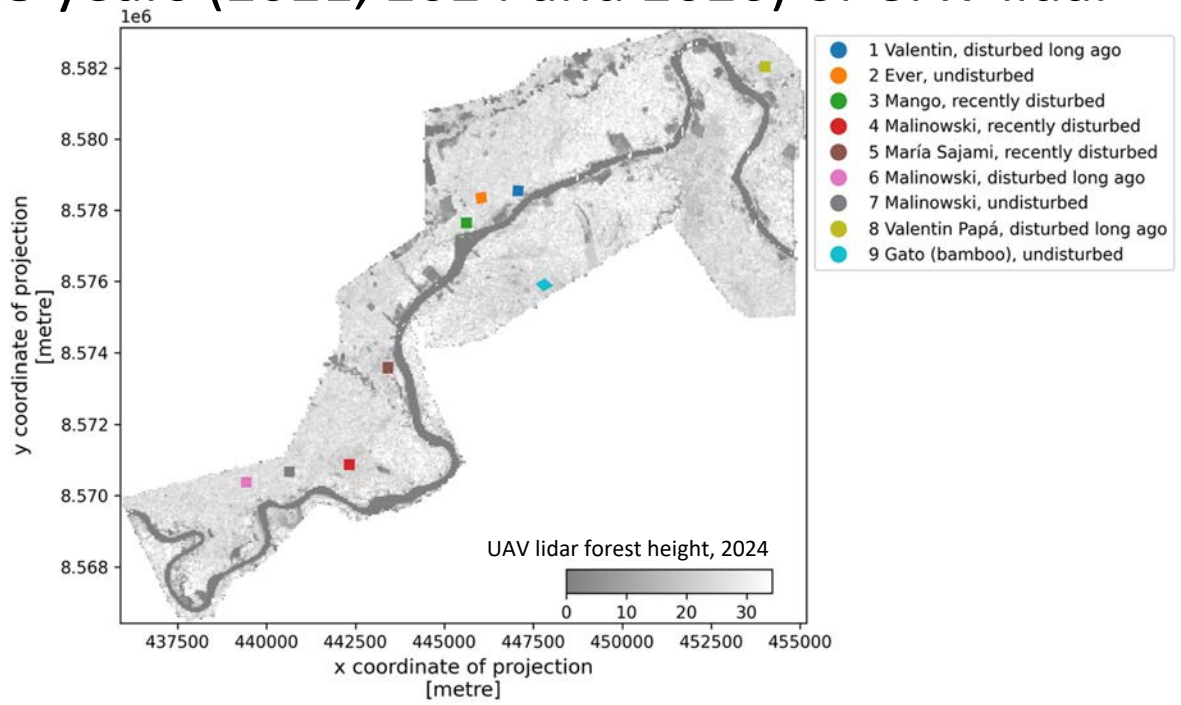


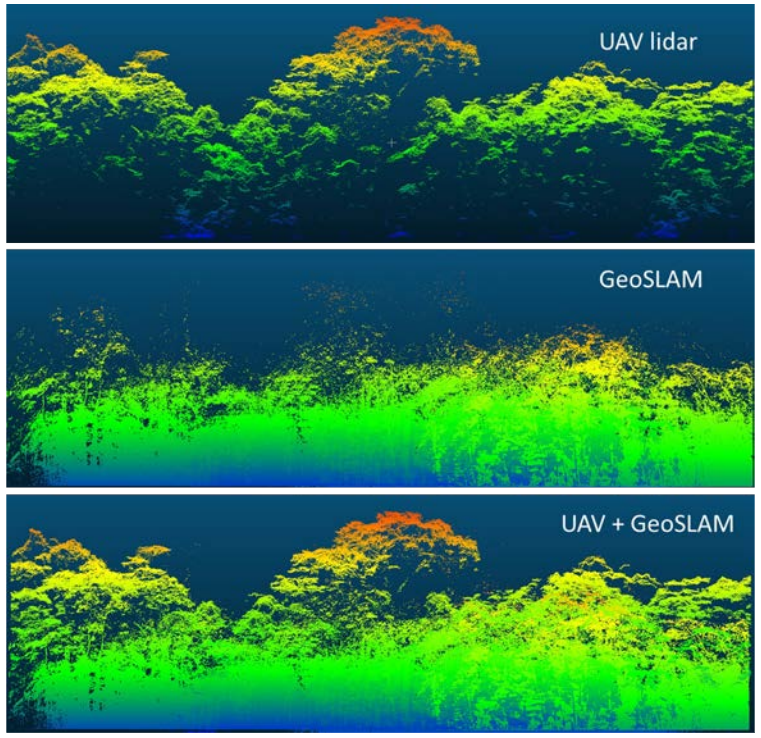
DTM in Mondah, Gabon



Focus #1: In situ campaigns

Completed campaigns in Tambopata, Peru (TU Dresden and Wilderness International)
9 hectares plots of inventory with forest disturbance gradient (2026)
6 hectares plots of hand-held laser scanning with GeoSLAM (September 2026)
3 years (2021, 2024 and 2026) of UAV lidar

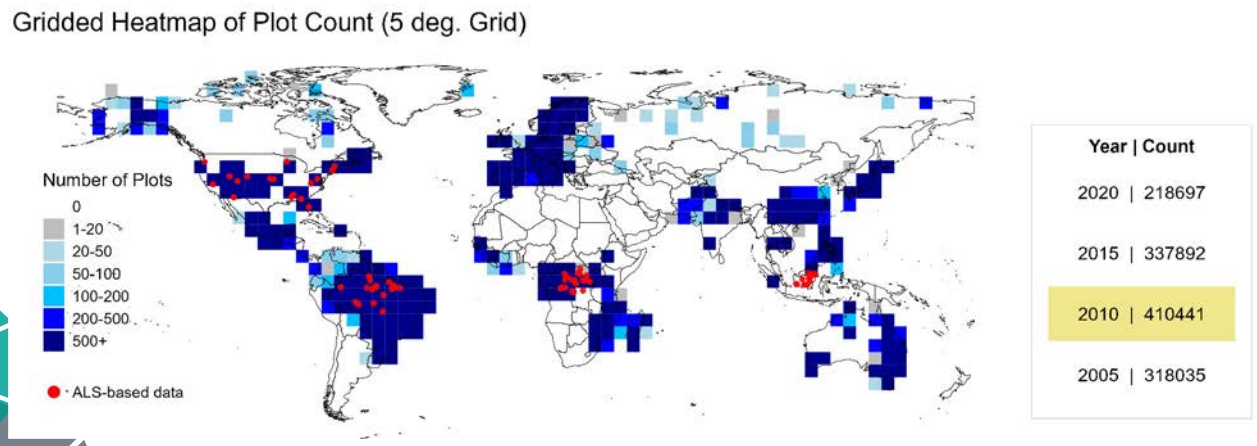


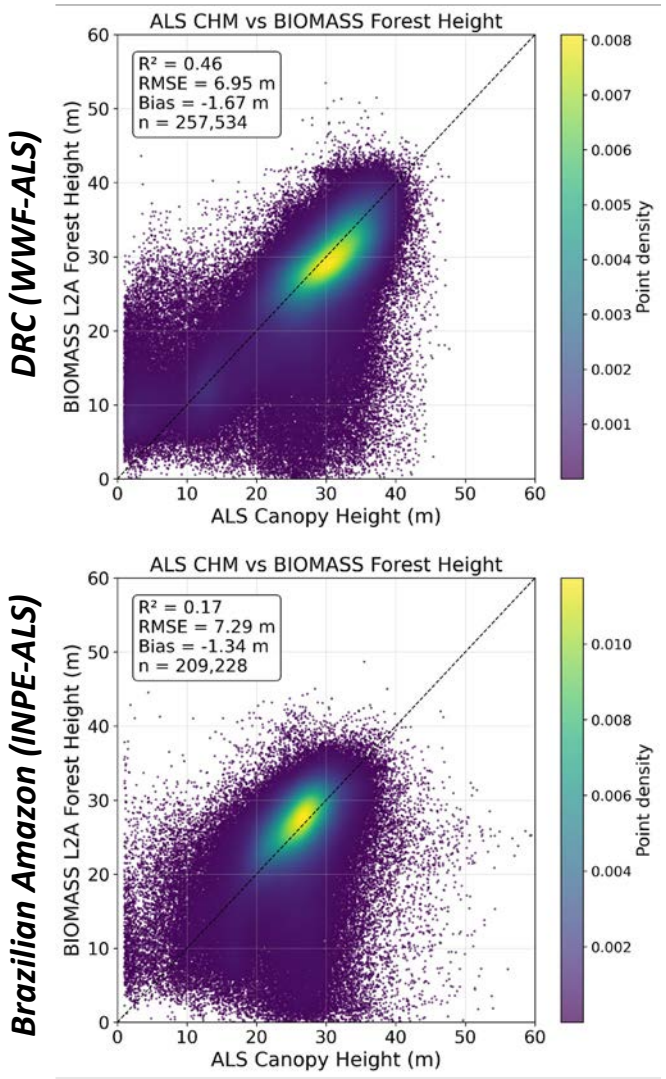


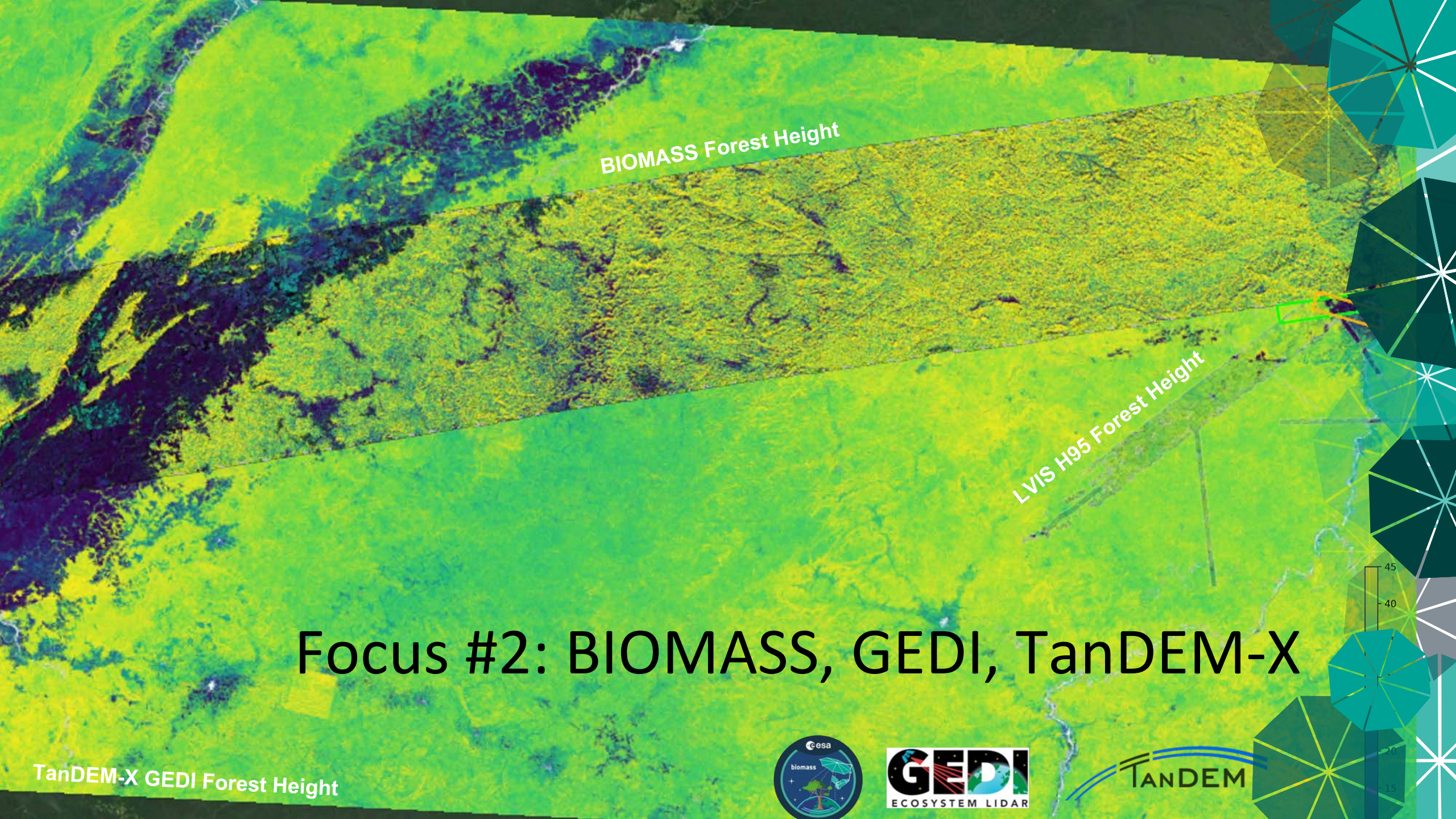
Focus #2: Compilation and comparison with existing reference data

Beyond GEO-TREES, a wide range of existing reference data (NFI, research plots, ALS) that can be used to assess BIOMASS products

***leveraging and updating AGBref** (Araza et al. 2026)*

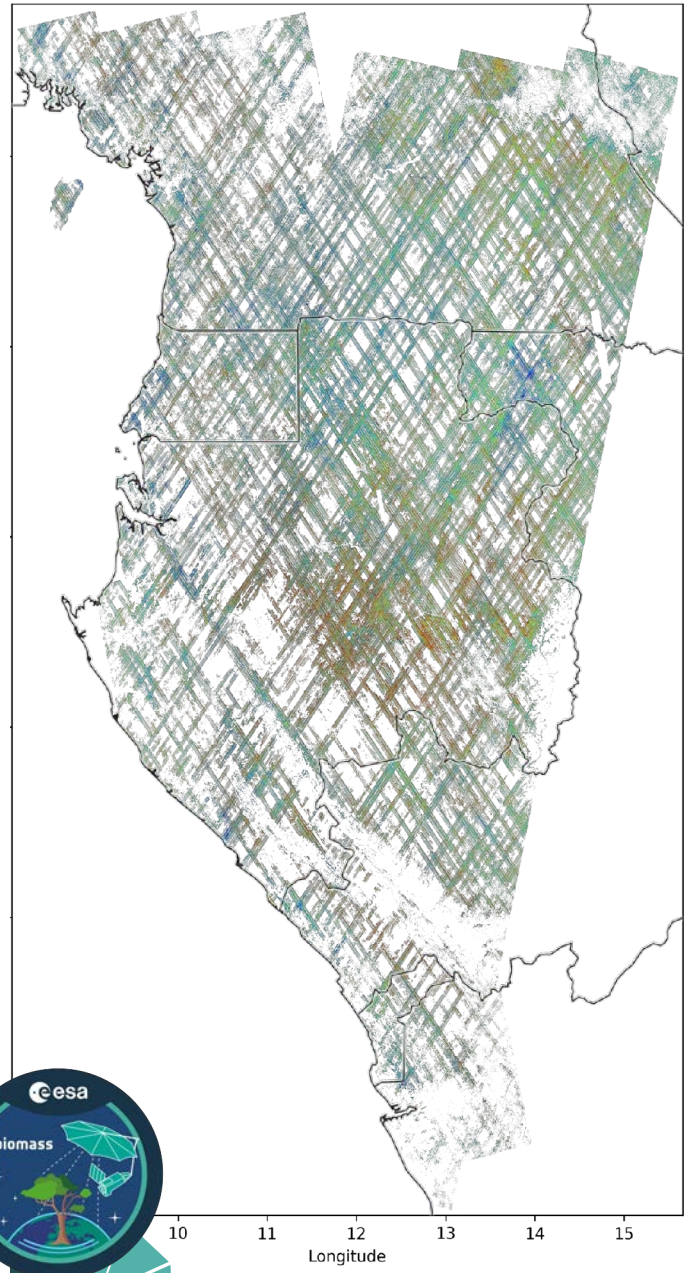




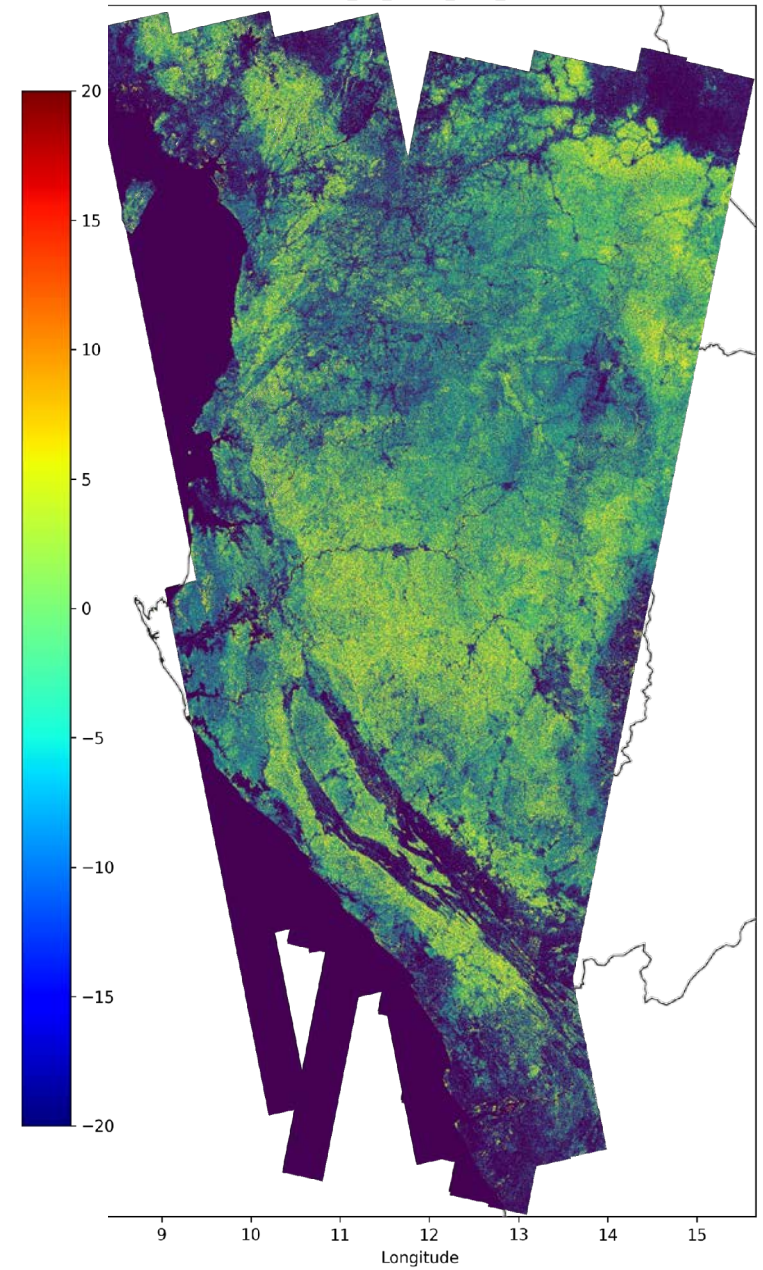


Focus #2: BIOMASS, GEDI, TanDEM-X

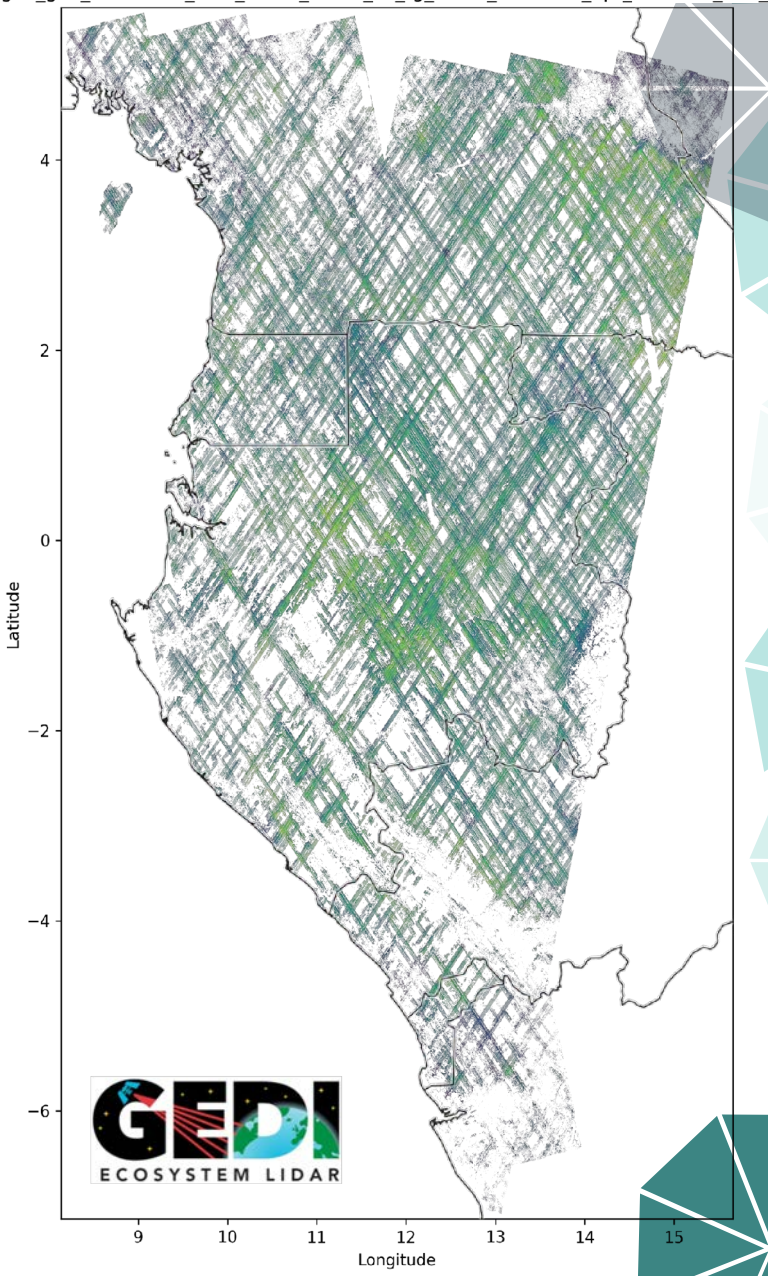
Forest Height Error vs. GEDI



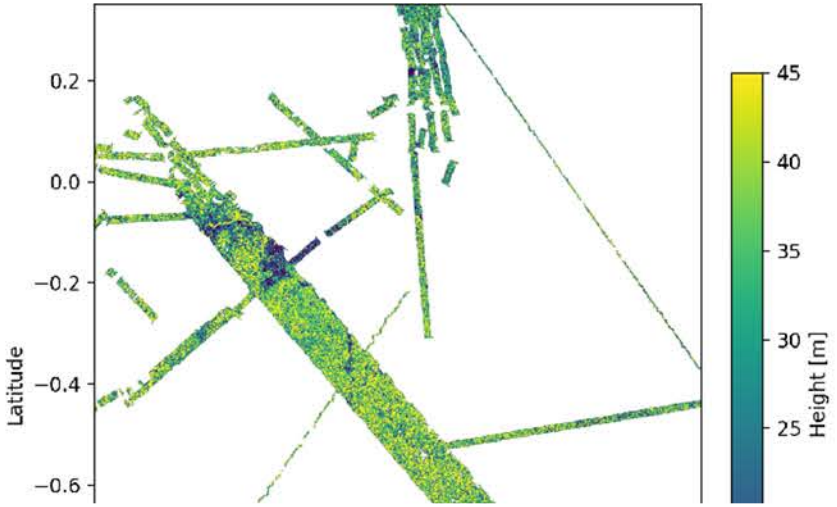
Forest Height



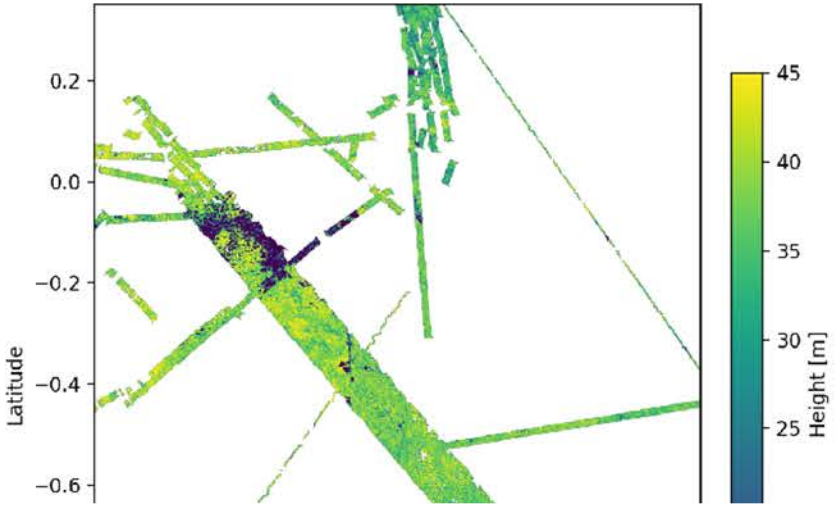
GEDI RH95



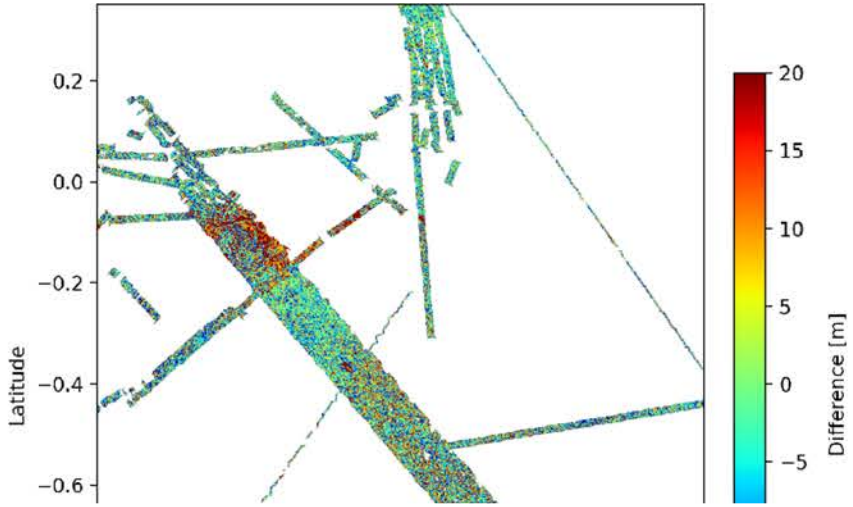
Forest Height BIOMASS

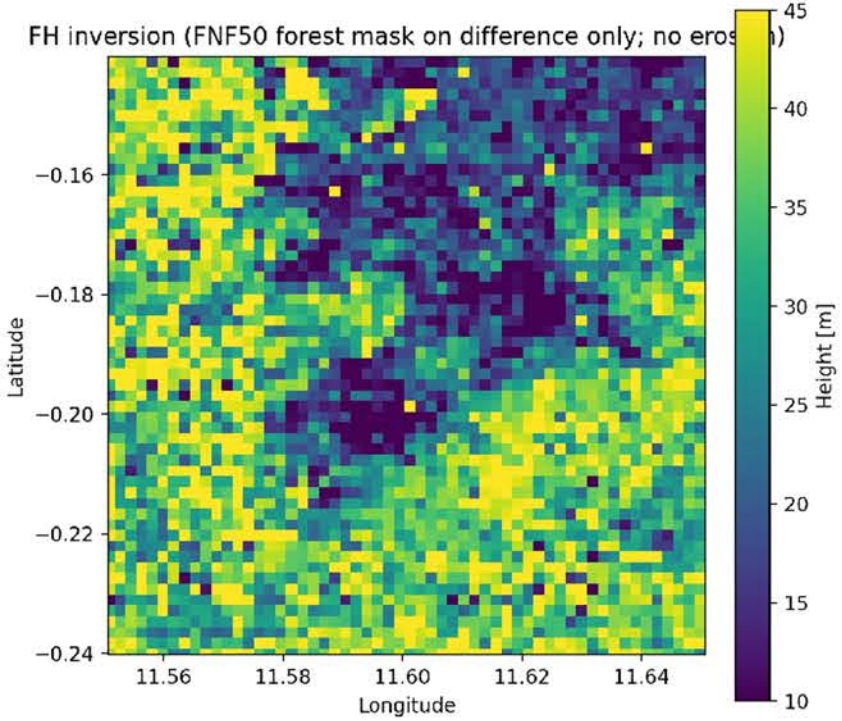


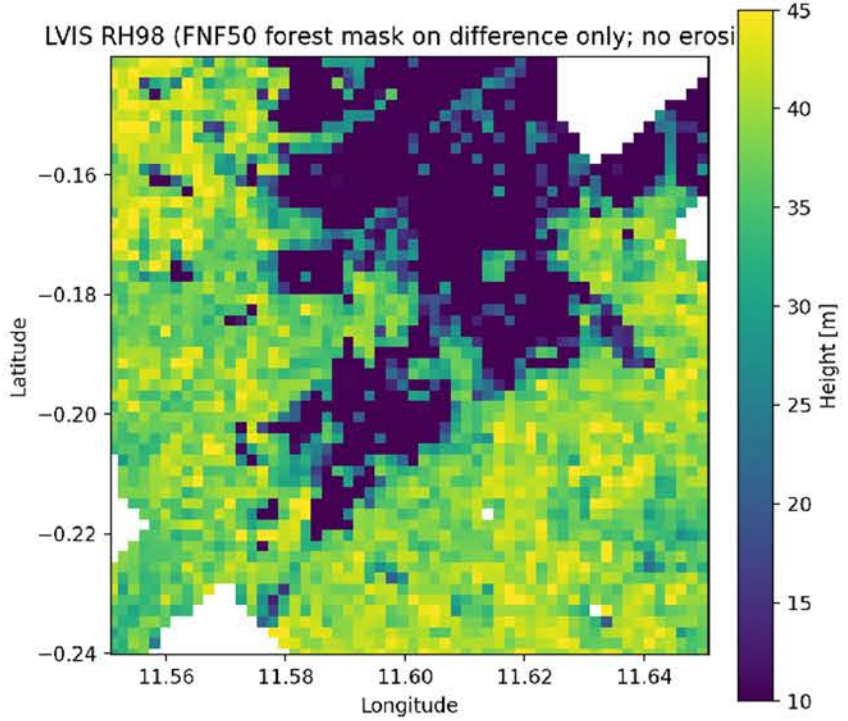
Forest Height LVIS

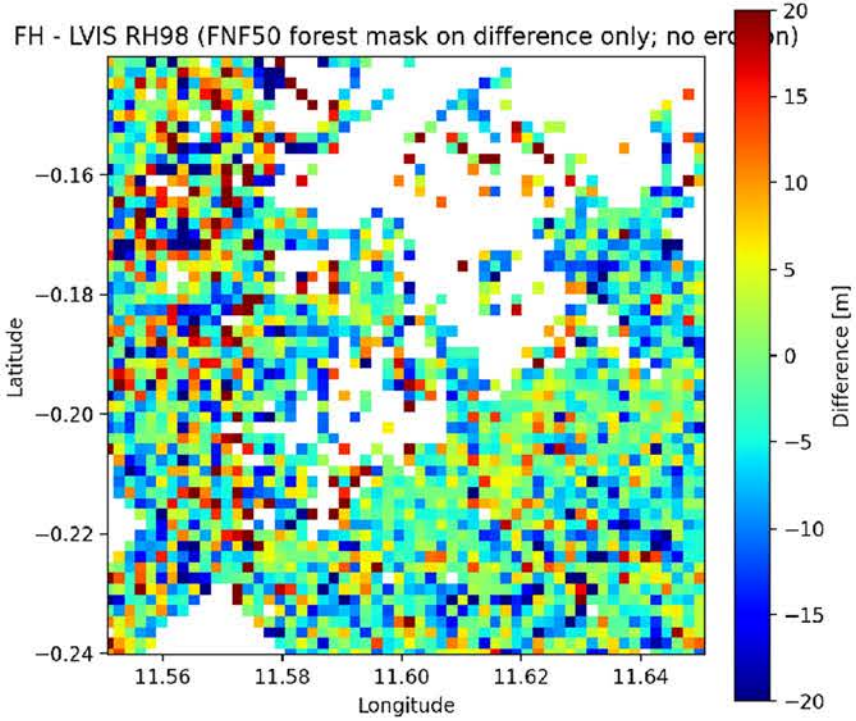


Forest Height Error



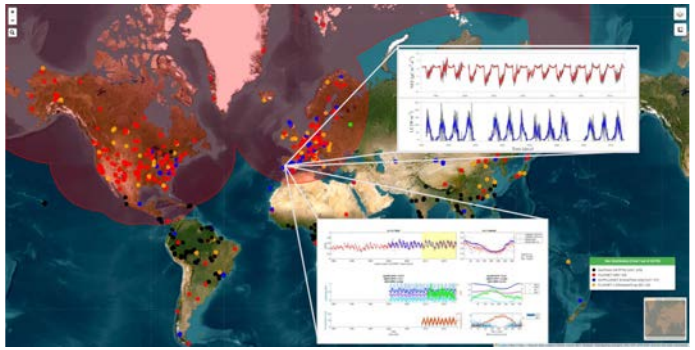




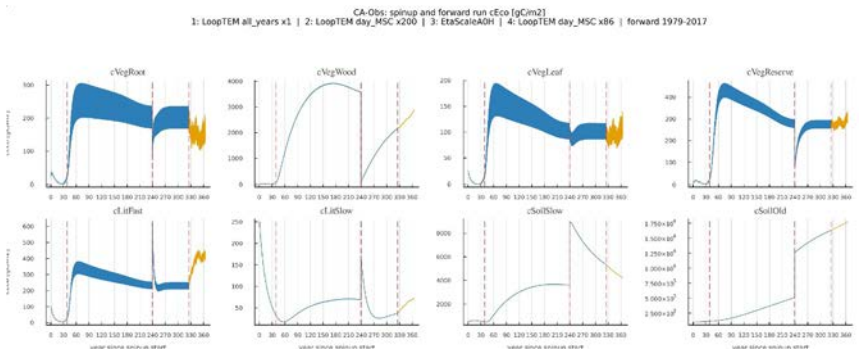


Focus #3: BIOMASS Carbon Cycle

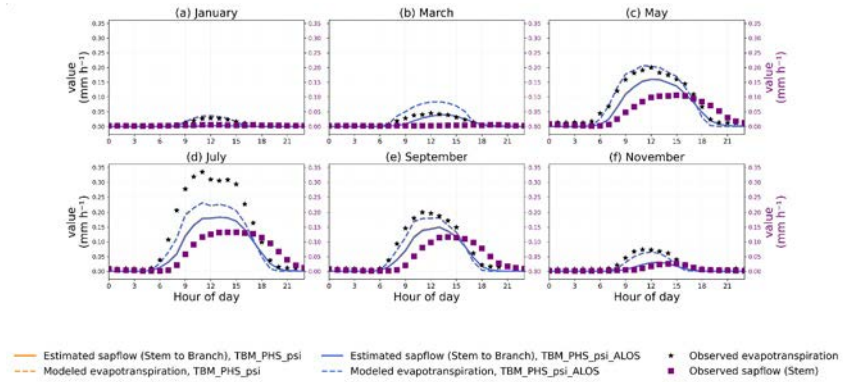
- Data collection



- Long term dynamics



- C-H2O flux dynamics



Focus #4: Coordination

- Communication & Dissemination

  https://www.eebiomass.org/en/decalval

- Project Management
- Reporting



Kick-Off Meeting in Dresden, December 2025

Thank you!